COHERENT MODULES ON A LOCALLY NOETHERIAN STACK FORM AN ABELIAN CATEGORY

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Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file stacks-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted span cites the lemmas recorded in assembly.external_references with canonical tags as navigation pointers.

Allowed standard prerequisites (editorial).

- QCoh rules for kernels and cokernels. Cohomology of Algebraic Stacks, Remark stacks-cohomology-remark-QCoh-abelian. Prescription agreement.
- Four coherent characterizations. Cohomology of Algebraic Stacks, Lemma stacks-cohomology-lemma-coherent-Noetherian. Coherence check on the presentation.
- Adjunction defect is parasitic. Cohomology of Algebraic Stacks, Lemma stacks-cohomology-lemma-adjoint-kernel-parasitic (2). f̂*Q(Ker) = f̂*Ker.

Source reference. Original remark 0GQL: sources/stacks-chow/tags.txt, source lines 20866–20866. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 0GR5: sources/stacks-chow/tags.txt, source lines 20885–20885. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 0774: sources/stacks-chow/tags.txt, source lines 9189–9189. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 0779: sources/stacks-chow/tags.txt, source lines 9194–9194. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let $\mathcal{X}$ be a locally Noetherian algebraic stack. The category of coherent $\mathcal{O}_\mathcal{X}$-modules is abelian. If $\varphi : \mathcal{F} \to \mathcal{G}$ is a map of coherent $\mathcal{O}_\mathcal{X}$-modules, then*

1. *the cokernel* $\mathrm{Coker}(\varphi)$ *computed in Mod$(\mathcal{O}_\mathcal{X})$ is a coherent $\mathcal{O}_\mathcal{X}$-module,*
2. *the image* $\mathrm{Im}(\varphi)$ *computed in Mod$(\mathcal{O}_\mathcal{X})$ is a coherent $\mathcal{O}_\mathcal{X}$-module, and*
3. *the kernel* $\mathrm{Ker}(\varphi)$ *computed in Mod$(\mathcal{O}_\mathcal{X})$ may not be coherent, but it is in $LQCoh^{fbc}(\mathcal{O}_\mathcal{X})$ and $Q(\mathrm{Ker}(\varphi))$ is coherent and is the kernel of φ in Coh$(\mathcal{O}_\mathcal{X})$.*

The inclusion functor Coh$(\mathcal{O}_\mathcal{X}) \to QCoh(\mathcal{O}_\mathcal{X})$ is exact.

Proof. The rules given for taking kernels, images, and cokernels in Coh$(\mathcal{O}_\mathcal{X})$ agree with the prescription for quasi-coherent modules in Remark 0GQL. Hence the lemma will follow if we can show that the quasi-coherent modules $\mathrm{Coker}(\varphi)$, $\mathrm{Im}(\varphi)$, and $Q(\mathrm{Ker}(\varphi))$ are coherent. By Lemma 0GR5 it suffices to prove this after

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restricting to $U_{étale}$ for some surjective smooth morphism $f : U \to \mathcal{X}$. The functor $\mathcal{F} \mapsto f^*\mathcal{F}|_{U_{étale}}$ is exact. Hence $f^* \operatorname{Coker}(\varphi)$ and $f^* \operatorname{Im}(\varphi)$ are the cokernel and image of a map between coherent $\mathcal{O}_U$-modules hence coherent as desired. The functor $\mathcal{F} \mapsto f^*\mathcal{F}|_{U_{étale}}$ kills parasitic modules by Lemma 0774. Hence $f^*Q(\operatorname{Ker}(\varphi))|_{U_{étale}} = f^* \operatorname{Ker}(\varphi)|_{U_{étale}}$ by part (2) of Lemma 0779. Thus we conclude that $Q(\operatorname{Ker}(\varphi))$ is coherent in the same way. □